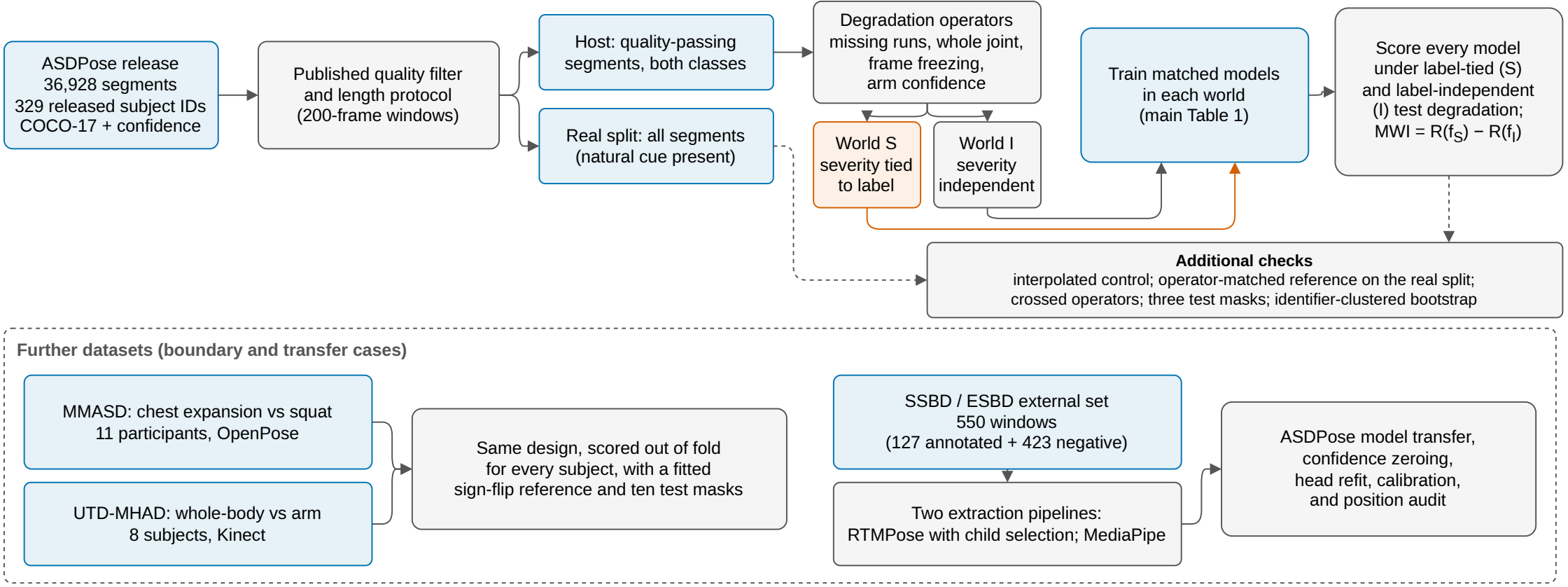
ASDPose release
36,928 segments
329 released subject IDs
COCO-17 + confidence
Published quality filter
and length protocol
(200-frame windows)
Host: quality-passing
segments, both classes
Real split: all segments
(natural cue present)
Degradation operators
missing runs, whole joint,
frame freezing,
arm confidence
World S
severity tied
to label
World I
severity
independent
Train matched models
in each world
(main Table 1)
Score every model
under label-tied (S)
and label-independent
(I) test degradation;
MWI = R(f_S) − R(f_I)
Additional checks
interpolated control; operator-matched reference on the real split;
crossed operators; three test masks; identifier-clustered bootstrap
Further datasets (boundary and transfer cases)
MMASD: chest expansion vs squat
11 participants, OpenPose
UTD-MHAD: whole-body vs arm
8 subjects, Kinect
Same design, scored out of fold
for every subject, with a fitted
sign-flip reference and ten test masks
SSBD / ESBD external set
550 windows
(127 annotated + 423 negative)
Two extraction pipelines:
RTMPose with child selection; MediaPipe
ASDPose model transfer,
confidence zeroing,
head refit, calibration,
and position audit